

2. The power  $P$  dissipated in a resistor of resistance  $R$  is calculated using the expression

$$P = \frac{V^2}{R}$$

where  $V$  is the potential difference across the resistor.

In an experiment to determine  $P$ ,  $R$  is measured to within a percentage uncertainty of 2%.

What is the maximum percentage uncertainty in the measurement of  $V$  such that  $P$  can be determined with a maximum uncertainty of  $\pm 5\%$ ?

**A** 1.5%

**B** 3.0%

**C** 3.5%

**D** 7%